# RFC OTA-Based LDO Voltage Regulator — UMC 180 nm CMOS

**M.Tech Thesis Project | NIT Silchar | Karnati Nagaraju**

A comparative design and simulation study of two Low-Dropout (LDO) voltage regulators using the Recycling Folded Cascode (RFC) OTA as the error amplifier, implemented in UMC 180 nm CMOS technology using Cadence Virtuoso/Spectre.

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## Overview

LDO regulators for battery-powered IoT sensor nodes face a fundamental three-way conflict: output voltage precision, loop stability, and quiescent current consumption — all coupled through the error amplifier transconductance and bias current.

This work proposes a **High-Swing RFC OTA with self-biased cascode architecture (Design 2)** and benchmarks it against a conventional RFC OTA baseline (Design 1). The key innovation is a self-biased cascode current mirror that simultaneously improves output voltage swing (~120 mV recovery), provides automatic PVT compensation, and eliminates parasitic bias current overhead — without external trimming.

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## Specifications

| Parameter | Value |
|---|---|
| Technology | UMC 180 nm CMOS (1P6M) |
| Supply voltage (V_DD) | 1.8 V |
| Regulated output (V_out) | 1.2 V |
| Max load current | 50 mA |
| Load capacitance | 1 µF |
| Feedback resistors | R1 = 50 kΩ, R2 = 100 kΩ |
| Pass transistor | W/L = 500 µm / 1 µm (PMOS) |
| Reference voltage | 0.6 V (ideal) |
| Simulator | Cadence Spectre (BSIM3) |

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## Design Comparison

| Parameter | Design 1 (Baseline RFC) | Design 2 (HS-RFC) | Change |
|---|---|---|---|
| Phase margin (TT, nominal) | 61.49° | **87.95°** | +43% |
| DC open-loop gain | 71.95 dB | 68.45 dB | −3.5 dB |
| GBW | 7.61 MHz | 2.51 MHz | −3× |
| Load regulation | 19.75 μV/mA | **7.98 μV/mA** | −60% |
| PSRR @ 1 kHz | −41.57 dB | **−46.62 dB** | +5.1 dB |
| Quiescent current | 66.32 μA | 72.46 μA | +9.2% |
| Settling time | 0.73 μs | 8.06 μs | +11× |
| Transient undershoot | 29 mV | 67 mV | — |
| Current efficiency | 99.87% | 99.86% | ≈same |
| Transistor sizing | Strong inversion | Moderate inversion (gm/ID = 17–19 $V^{-1}$) | — |
| Biasing | Static external | Self-biased cascode | — |

Design 2 achieves the **highest phase margin and lowest load regulation** among 9 published LDO designs compared in the thesis (Table 5.8).

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## PVT Robustness (Design 2)

Full 27-point factorial analysis: 3 process corners (SS, TT, FF) × 3 supply voltages (1.62, 1.80, 1.98 V) × 3 temperatures (−40°C, 27°C, 125°C).

| Metric | Result |
|---|---|
| Phase margin variation (all 27 PVT) | **0.50°** |
| Minimum phase margin (worst-case) | 86.52° @ SS, 1.62 V, −40°C |
| Worst-case load regulation | 48.42 μV/mA @ 125°C |
| Best-case PSRR @ 1 kHz | −48.71 dB |

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## Schematics

### LDO System Architecture

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### Design 1 — Baseline RFC OTA (Static External Biasing)

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### Design 2 — Proposed High-Swing RFC OTA (Self-Biased Cascode)

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## Simulation Results

### Frequency Response — Design 1 vs Design 2

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### PSRR — Design 1 vs Design 2

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### Transient Response — Load Step 1 mA → 50 mA

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### Stability (Full Load) — Process Corners

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### Stability (Full Load) — Temperature Variation

Show Image

### PSRR — Process Corners

Show Image

### PSRR — Temperature Variation

Show Image

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## Key Technical Contributions

1. **Novel self-biased high-swing cascode biasing for RFC OTA-based LDOs** — cascode gate voltages self-generated from the circuit's own bias network using diode-connected devices.

Recovers ~120 mV output swing, provides automatic PVT tracking, eliminates parasitic bias current from external resistors.

2. **First documented gm/ID sizing procedure for RFC OTA in LDO context** — systematic step-by-step procedure sizing RFC OTA transistors in moderate inversion using PDK lookup tables (UMC 180 nm). Achieves 40–50% reduction in drain current at the same transconductance target vs. strong inversion.
3. **Comprehensive 27-point PVT validation** — full factorial PVT analysis demonstrating 0.50° total phase margin variation, considerably more thorough than typical 3-corner reports in published literature.
4. **Quantitative speed-stability trade-off characterisation** — two RFC OTA variants occupying different regions of the speed-stability-power design space compared in a structured manner.

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## Publication

> **Karnati Nagaraju**, Sourav Chanda, and Dr. Bijit Choudhuri, "Design of a High-Precision, High-Stability Low-Dropout Regulator Using High-Swing Recycling Folded Cascode OTA," in *Proc. International Conference on Emerging Trends in Technology, Engineering, Computing and Networking (E2TECN 2026)*, SVNIT Surat, April 2026. **Published in Springer Lecture Notes in Electrical Engineering (LNEE). [Presented]**

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## Limitations

- **Simulation only** — all results from schematic-level Cadence Spectre simulation. No physical layout, post-layout extraction, or silicon fabrication.
- **Ideal voltage reference** — Vref implemented as a fixed 0.6 V ideal source. A real bandgap reference would add temperature drift and noise.
- **Proprietary PDK** — UMC 180 nm CMOS PDK (BSIM3 models) is not distributable. Netlists are not included.
- **No startup circuit** — design assumes bias currents are correctly established at power-on.
- **Model-to-hardware correlation** — errors of 5–10% are typical for circuits of this complexity.

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## Repository Structure

```
RFC-OTA-LDO-UMC180nm/
├── README.md
├── docs/
│   ├── thesis.pdf                    # M.Tech thesis (NIT Silchar, 2025-26)
```

```
│   └── E2TECN_Manuscript_Nagaraju.pdf  # Conference paper (Springer LNEE)
├── schematics/
│   ├── LDO_SCHEMATIC_FINAL1.jpg        # System-level LDO architecture
│   ├── BASELINE_OTA_final.jpg          # Design 1: Conventional RFC OTA
│   └── PROPOSED_OTA_FINAL.jpg          # Design 2: High-Swing RFC OTA
└── results/
    ├── gain__GBW__PM_of_design1___2.jpeg
    ├── psrr_of_both_designs_.png
    ├── transient_final_v7_1_.png
    ├── phase_full_process.png
    ├── phase_full_temp.png
    ├── phase_light_process.png
    ├── phase_light_temp.png
    ├── phasemargin__full_voltage.png
    ├── phasemargin_light_voltage.png
    ├── psrr_process.png
    ├── psrr_temp.png
    └── psrr_voltage.png
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